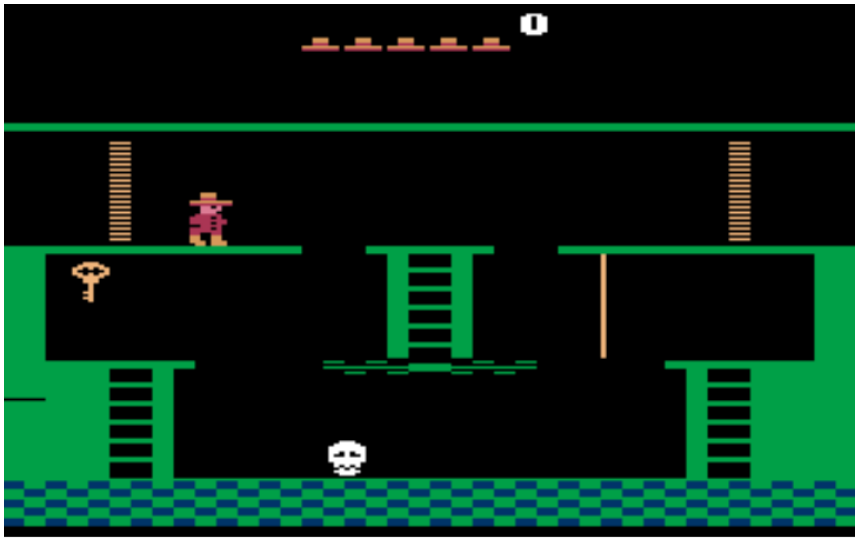
Homework 6: Policy Gradient Reinforcement Learning

CS 2470

Due November 22nd, 2019 at 11:59pm

1 CS2470-only Questions

1. RL agents struggle when the MDP they are trying to solve requires a long sequence of coordinated actions before any reward signal is received. This motivates the practice of "reward shaping," or trying to design reward functions that somehow gradually reward an agent for making progress toward its ultimate goal. A classic example of this phenomenon occurs in the Atari game Montezuma's Revenge. The included figure shows the starting screen of the first level of this game.



   (a) Assume that the agent receives a reward of 1 (i.e. a score increment of 1) when it retrieves the key located on the left part of the screen. In order to get to this key, the agent must learn to jump across platform gaps, climb down a latter, jump onto a hanging rope, fall off the rope, avoid the moving skull enemy, and climb up the final ladder to the key. Design a reward function that will encourage the agent to get

to the key as quickly as possible, while avoiding death (assume you have access to the complete state of the game).

Answer: One of the design of reward function could be the distance between the agent and the key. When the distance could approach key closer, the higher rewards the agent could receive. The agent could receive higher rewards if he could jump the death trap and climb the ladder.

(b) For an actual RL agent trying to play Montezuma's revenge, the situation is more dire: the only input the agent can see are pixels, and obtaining the key actually does not confer any reward. Rather, reward is only obtained if the agent obtains the key, moves to a different screen to the left or right of the current screen, and uses the key to unlock something. Can you think of possible reward functions that might encourage this behavior (in this case, you only have access to the raw game pixels).

Answer: We could design a reward for the agent given the condition of with or without the key. If the agent has not reached the key, we could design the reward function as the pixel distance between the agent and the key. If the agent has already reached the key, we could design the reward function as the distance between the agent and the door.

(c) If your strategy from part 2 was specific to Montezuma's Revenge: now can you think of ways it might be generalized to apply to any Atari game (i.e. any MDP where the inputs are pixels, the actions are discrete controller inputs, and the reward is game score)?

Answer: We could always divide the ultimate goal into a series of separate sub-goals or steps. Using the strategies in (b), the reward function could guide the agent to finish a series of moves to the wanted pixels, and hence, finishing these sub-goals one by one. The ultimate goal would be automatically finished after finishing each separate goals.

2. The objective function for policy gradients is defined as:

$$J(\theta) = \mathbb{E}[\sum_{t=0}^{T-1} r_{t+1}]$$

Derive the policy gradient i.e $\nabla_\theta J(\theta)$

Hints:

(a) *Rewrite the expectation using your policy* π_θ

(b) *Write out the objective with respect to an arbitrary trajectory*

(c) *Recall* $\frac{d}{dx}\log\left(f(x)\right) = \frac{f'(x)}{f(x)}$

Proof:

$$J(\theta) = \mathbb{E}[\Sigma_{t=0}^{T-1} r_{t+1}|\pi_\theta] = \Sigma_{t=i}^{t-1} P(s_t, a_t|\tau) r_{r+1} \tag{1}$$

where τ is the trajectory and $\pi(\theta)$ is the policy function. Take the derivative with respect to θ for J_θ, we get

$$\nabla_\theta J(\theta) = \Sigma_{t=i}^{T-1} \nabla_\theta P(s_t, a_t|\tau) r_{t+1}, \tag{2}$$

$$= \Sigma_{t=i}^{T-1} P(s_t, a_t|\tau) \frac{\nabla_\theta P(s_t, a_t|\tau)}{P(s_t, a_t|\tau)} r_{t+1}, \tag{3}$$

$$= \Sigma_{t=i}^{T-1} P(s_t, a_t|\tau) \nabla_\theta \log P(s_t, a_t|\tau) r_{t+1}, \tag{4}$$

$$= \mathbb{E}[\Sigma_{t=i}^{T-1} \nabla_\theta \log P(s_t, a_t|\tau) r_{t+1}], \tag{5}$$

$$\tag{6}$$

Decompose $\nabla_\theta \log P(s_t, a_t|\tau)$ as:

$$\nabla_\theta \log P(s_t, a_t|\tau) = \nabla_\theta \log P(s_0) + \nabla_\theta \log \pi_\theta(a_1|s_0) + \tag{7}$$

$$\nabla_\theta \log P(s_1|s_0, a_0) + \nabla_\theta \log \pi_\theta(a_2|s_1) + \tag{8}$$

$$\nabla_\theta \log P(s_2|s_1, a_1) + \nabla_\theta \log \pi_\theta(a_3|s_2) + ... \tag{9}$$

$$\nabla_\theta \log P(s_t|s_{t-1}, a_{t-1}) + \nabla_\theta \log \pi_\theta(a_t|s_{t-1}) \tag{10}$$

$$= \nabla_\theta \log \pi_\theta(a_1|s_0) + \nabla_\theta \log \pi_\theta(a_2|s_1) + \nabla_\theta \log \pi_\theta(a_3|s_2) + ... + \nabla_\theta \log \pi_\theta(a_t|s_{t-1}) \tag{11}$$

$$= \Sigma_{t'=0}^{t} \nabla_\theta \log \pi_\theta(a_{t'}|s_{t'}) \tag{12}$$

Plugging this back original equation:

$$\nabla_\theta J(\theta) = \mathbb{E}[\Sigma_{t=i}^{T-1} \nabla_\theta \log P(s_t, a_t|\tau) r_{t+1}], \tag{13}$$

$$= \Sigma_{t=i}^{T-1} r_{t+1} \Sigma_{t'=0}^{t} \nabla_\theta \log \pi_\theta(a_{t'}|s_{t'}), \tag{14}$$

$$= \nabla_\theta \log \pi_\theta(a_0|s_0)(r_1 + r_2 + r_3 + ...r_T) + \nabla_\theta \log \pi_\theta(a_1|s_1)(r_2 + r_2 + r_3 + ...r_T) + ... \tag{15}$$

$$+\nabla_\theta \log \pi_\theta(a_{T-1}|s_{T-1})(r_T) \tag{16}$$

$$= \nabla_\theta \log \pi_\theta(a_t|s_t)(\Sigma_{t'=t+1}^{T} r_T) \tag{17}$$

$$= \nabla_\theta \log \pi_\theta(a_t|s_t) D(s_t, a_t) \tag{18}$$